FINAL YEAR RESEARCH PROJECT
Review-1 Presentation

THRESHOLD & MARGIN OPTIMIZATION FOR ROBUST FACE VERIFICATION

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Presentation Outline

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  - Motivations
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- Proposed Methodology & Approach
- Dataset & Tools/Technologies
- Implementation Progress
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Introduction

THE OPPORTUNITY

Critical Domains Surveillance | Banking | Smart Cities | Access Control

Deep Learning Higher accuracy, speed and autonomous recognition

AI-Powered Security More autonomous and intelligent authentication systems

BUT

THE CHALLENGE

Adversarial Attacks Imperceptible perturbations fool even top models

Brittle Thresholds Fixed and adaptive thresholds both fail under real attacks

No Explainability Systems lack reasoning, validation and interpretability

We need an **intelligent, adaptive, and explainable** authentication framework — that is exactly what this work proposes.

Problem Statement

01 Face recognition systems are vulnerable to adversarial attacks

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02 Fixed thresholds fail; adaptive ones bypass-able by doppelganger attacks

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03 Systems focus on model accuracy, not decision-making quality

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04 No reasoning, validation, or interpretability in outputs

CONSEQUENCES

- False acceptance and false rejection of identities
- One-size-fits-all decisions cannot handle varying attack scenarios
- Lack of explainability erodes trust in authentication decisions
- Critical security risks in banking, access control and surveillance

Need: An intelligent, adaptive, and explainable framework that dynamically assesses adversarial risks and makes robust authentication decisions.

Motivations

EXISTING GAPS		OUR SOLUTION PILLARS
Biometric systems are widely deployed but poorly defended		**Intelligent** Dynamic adversarial risk assessment at decision time
Threshold-based approaches unreliable in complex real-world scenarios		**Adaptive** Margin and threshold optimization for real-world attack scenarios
Decisions made directly from model outputs without validation	⇒	**Explainable** Transparent reasoning and validation in every authentication step
Lack of explainability reduces trust in security decisions		**Robust** Reliable authentication even against sophisticated doppelganger attacks

Each gap above directly motivates one or more pillars of the proposed framework

Literature Review

Table 1:Summary of Existing Approaches and Research Limitations

S. No.	Author & Year	Methods/Techniques Used	Dataset Used	Research Outcomes	Findings
1	Park et al. (2024)	Attack surfaces (7), scenarios (12), G-T-C framework, dependency-aware modeling, adversarial simulation (FGSM, Square, Witches' Brew, GMI, BREP-MI), logic corruption.	FaceForensics++, CASIA-WebFace, custom test sets	Risk metrics: vulnerability (ASR), difficulty (info), defense availability; risk grading (1–5 aggregated).	Scenario-based multi-metric risk framework with attack simulation, but limited to face-based evaluation, equal-weight assumptions, and lacking real-world validation factors.
2	Verheyen et al. (2023)	Adaptive thresholds (system/template), identity-level thresholding, PGD attacks, template similarity (intra/inter), threshold exploitation, similarity-based sanitization.	VGGFace2, CASIA-WebFace (pretraining)	Static: CIR 72%, FAR 30%, FRR 28%; template-adaptive: CIR 68%, FAR 32%, FRR 12%, CVR 88%; system-adaptive: CIR 42%, FAR 58%, FRR 40%; identity-adaptive: CIR 82–85%, FAR 8%, FRR 10%, CVR 85%; doppelgänger: CIR 20%, CVR 40%, FRR up to 80%.	Adaptive threshold vulnerability via doppelgänger attacks with identity-level mitigation, but reliant on white-box settings, limited to digital attacks, and lacking real-world validation.

Literature Review cont.

Table 1:Summary of Existing Approaches and Research Limitations

S. No.	Author & Year	Methods/Techniques Used	Dataset Used	Research Outcomes	Findings
3	Apap et al. (2024)	3-layer COSFIRE, Gabor filters, keypoint modeling, spatial encoding, rotation invariance, geometric fusion, similarity scoring, one-shot learning-free config, histogram detection.	VARIA, RIDB, IITD Palmprint	Retina (VARIA/RIDB): RR 100%; palmprint (IITD): RR 97.54% (100%), 97.62% (90%), 95.49% (80%); partial retina: 98.89%/100% (90%), 94.71%/94.20% (80%); adversarial detection: 100% (AUC 1), clean RR 100%.	Explainable learning-free COSFIRE framework for retina/palmprint with strong robustness and partial matching, but limited to shape-based patterns and lacking scalability and real-world validation.
4	Sureshkumar et al. (2025)	CNN+ResNet fusion, multi-level features, concatenation, binary mapping, spoof detection, ElGamal+AES crypto, Ed25519, PCA, adaptive learning.	CelebA, Replay-Attack, MSU-MFSD	Model (CelebA): Acc 97.1%, Loss 0.04, Prec 96.71%, Rec 97.10%, F1 96.38%; baselines: CNN 93.8/84.78/86.54, Deep CNN 91.0/88.67/88.12, CNN-2 95.0/88.40/79.09, AMS 94.5/93.86/94.48; ablation: CNN 95.6/EER 4.2/AUC 0.98, ResNet 96.0/3.8/0.985, CNN+ResNet 96.77/3.3/0.986, full 96.80/3.2/0.987; time 0.62 ms, crypto 41 ms.	CNN–ResNet with cryptographic biometric framework achieving high accuracy and spoof resistance, but limited by dataset bias, generalization issues, computational overhead, and lack of real-world validation.

Literature Review cont.

Table 1:Summary of Existing Approaches and Research Limitations

S. No.	Author & Year	Methods/Techniques Used	Dataset Used	Research Outcomes	Findings
5	Guo et al. (2025)	Pixel attacks (FGSM, PGD, MI/N-MI-FGSM), optimization, generative (GAN/VAE/diffusion), physical, latent, transferability, semantic manipulation, defense applications.	ImageNet, COCO, CIFAR-10, CIFAR-100, MNIST, synthetic (GAN/Diffusion)	ASR >90%, detection degradation 75%, physical ASR >80%, 3D ASR 85%, RFLA ASR 99%.	Comprehensive adversarial attack taxonomy and evolution across domains, highlighting dual threat-defense roles, but limited by efficiency issues, lack of standard evaluation, and insufficient real-world validation
6	Ashis Kumar Pati (2025)	RL/IRL/IL, MAS+swarm (ACO, PSO), neuro-symbolic, NLP/LLMs, knowledge graphs, CV+sensor fusion, planning (STRIPS, HTN), architectures (BDI, SOAR, ACT-R), simulation.	Organizational datasets (no fixed benchmark)	Time −34.2%, accuracy +7.7%, resource +13.6%; workforce risk: high 10.6%, moderate–high (women) 44.4%.	Comprehensive survey of agentic AI architectures and applications showing efficiency and adaptability gains, but lacking standardized benchmarks, scalability, and real-world validation.

Literature Review cont.

Table 1:Summary of Existing Approaches and Research Limitations

S. No.	Author & Year	Methods/Techniques Used	Dataset Used	Research Outcomes	Findings
7	Chen et al.(2026)	Multi-group debate (G, N=3), reasoning (IO, CCoT, DDCoT), iterative refinement (R=3), leader selection, logical audit, propagation, aggregation, fixed roles.	MM-Vet, ScienceQA	MM-Vet: single 57.47–63.49%, DMAD 64.54%, G-DMAD 73.07%, GLM G-DMAD 73.35%; ScienceQA: GLM 95.12%/93.88%, GPT 90.45%/89.80%; efficiency: tokens 3.73M→3.92M (+5%), accuracy +8.53%.	Introduced G-DMAD with state-of-the-art multi-group reasoning and efficiency optimization, but limited by fixed grouping, narrow benchmark evaluation, LLM dependency, and scaling overhead.
8	Li et al. (2025)	Multi-agent debate (6), specialist analysis (URL/HTML/content/brand), multi-round (R=1–10), consensus, judge decision, modular agents, prompt reasoning, parsing (BeautifulSoup), truncation.	Mendeley Phishing Dataset, TR-OP Dataset	PhishDebate: GPT-4o 96.50/94.97/98.2/96.56, GPT-4o-mini 93.90/90.57/98.00/94.14; Gemini 92.44/99.2/92.97; Qwen 85.60/84.84/FNR 19.4%; comparison (mini): single 67.0/74.69/4.7s, CoT 90.70/90.94/10.5s, debate 93.90/94.14/37.5s.	Multi-agent phishing detection framework improving accuracy and interpretability, but limited by LLM dependency, computational cost, and lack of adversarial robustness evaluation.

Literature Review cont.

Table 1:Summary of Existing Approaches and Research Limitations

S. No.	Author & Year	Methods/Techniques Used	Dataset Used	Research Outcomes	Findings
9	Alghamdi et al. (2024)	Multimodal (face+gait), ArcFace-ResNet34, DenseNet201, GEI, fusion (input/feature-SE/decision), enhancement (SE, attention), FGSM (ε=0.001–0.1), white-box impersonation, augmentation.	ORL, FEI, CASIA-B	Fusion: input ASR 13.23–31.61%, feature 84.19%, decision max 84.51%, avg 83.87%; clean baseline: Acc/Prec/Rec/F1 ~95–100%.	Multimodal biometric fusion improving robustness against adversarial attacks, but limited to FGSM, constrained modalities, and lacking real-world validation.
10	Lin et al. (2023)	Landmarks (68), superpixels, mask generation, mask-guided attacks, FGSM/BIM/PGD, ε-bounded iteration, CE+TV loss, white/black-box, physical simulation.	CASIA-WebFace (filtered subset)	FGSM: ASR 90.72%, 61.44% (mask), SSIM 0.5564, 0.9714; BIM: ASR 93.12%, 90.72%, SSIM 0.8523, 0.9830; PGD: ASR 100%, 90.56%, SSIM 0.8350, 0.9790.	Mask-guided adversarial attack improving imperceptibility and physical robustness, but with reduced attack success, environmental sensitivity, and limited scope.

Research Gap & Improvements

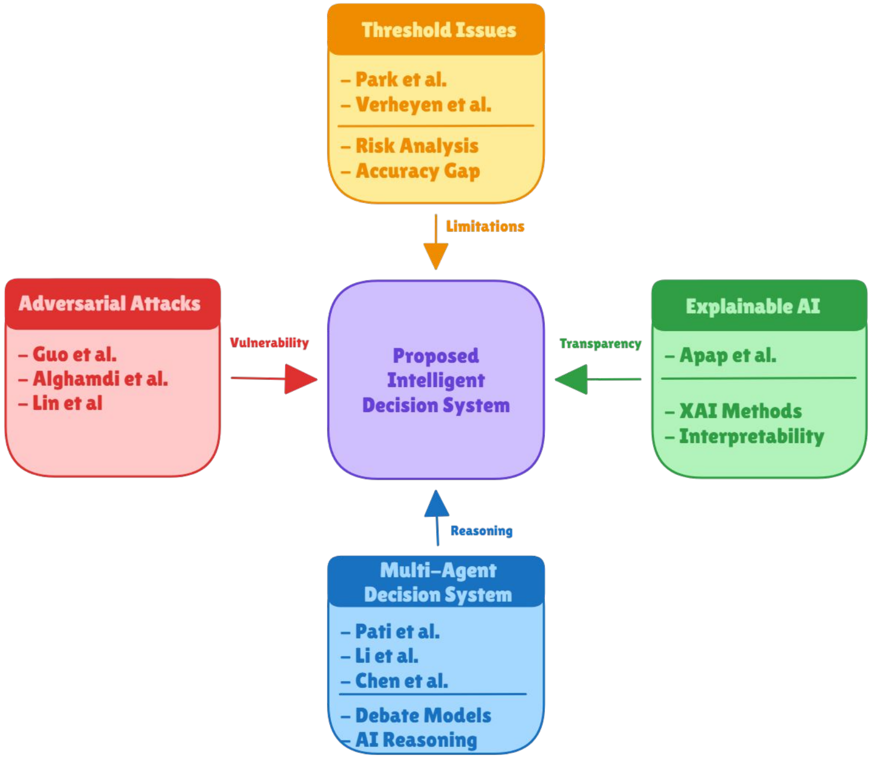

Fig. 1: Research gaps and Improvements filled by our Proposed model

Proposed Solution & Architecture

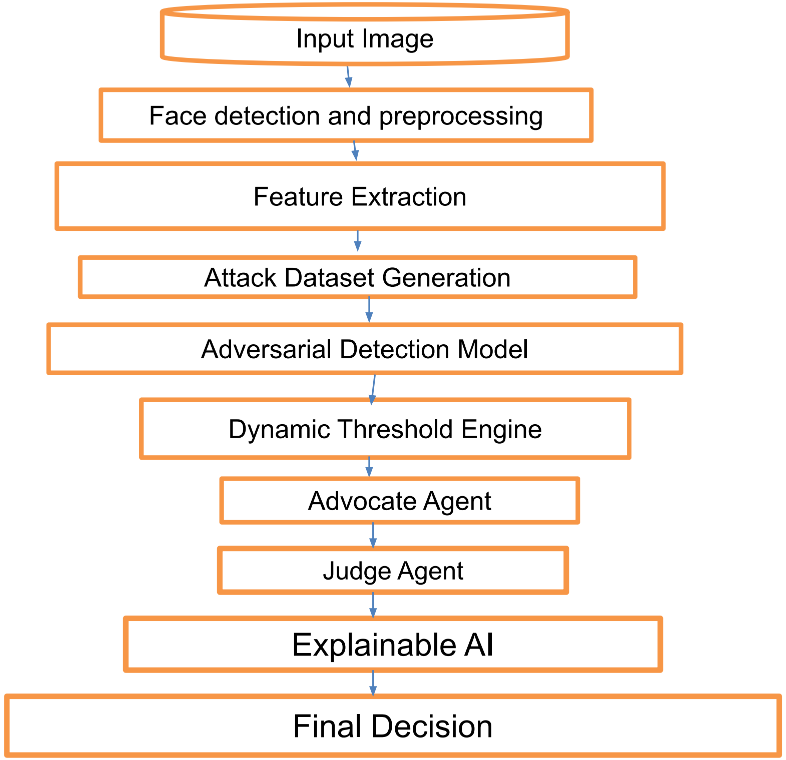

Fig. 2: Workflow of the Proposed Model

Proposed Solution & Architecture cont.

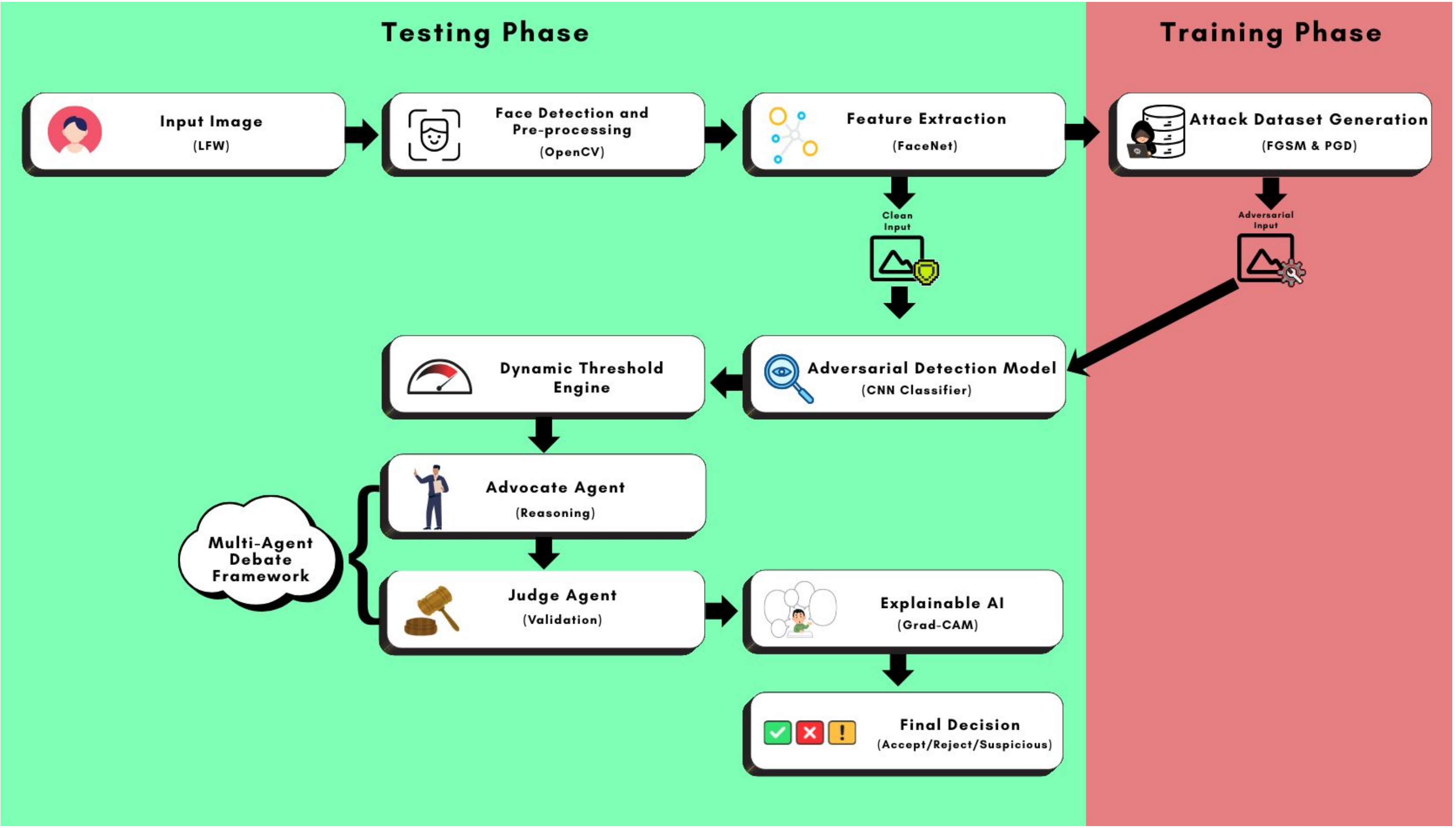

Fig. 3: Schematic Model Diagram of the Proposed Model

Dataset & Tools/Technologies

Datasets Used:

- CASIA-WebFace - 494,414 images, 10,575 identities
- Labeled Faces in the Wild (LFW) - 13,233 images, 5749 identities
- CelebFaces Attribute (CelebA) - 20,000 images, 10,000 identities
- Attack dataset (adversarial sample generation)

Tools/Technologies Used:

- CNN-based architectures (ResNet-18)
- Fast Gradient Signed Method (FGSM)
- Projected Gradient Descent (PGD)
- Large Language Models (LLMs)
- Multi-agent Debate Framework (MAD)
- Gradient-weighted Class Activation Mapping (Grad-CAM)
- Python
- TensorFlow/PyTorch
- Computer vision pipelines (OpenCV)

Progress in Implementation Plan & Methodology

1

2

Research paper procurement and selecting only what align with our objective

Exploring datasets, models & building proposed solution's architecture

Future Work Plan

3

Implementing backend, code formation, integrating necessary algorithm and datasets

4

Building the frontend and connecting it to the backend

5

Deploying the entire project as a Full-Stack application

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Thank you!